

Industrial Internship Report on

"File Organizer"

Prepared by

Shailesh Sharma

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was file Organizer. This 6-week internship report highlights your **Automated Python File Organizer** project developed with UCT:

- **Core Objective:** Eliminate digital storage clutter by automating file organization.
- **Your Role:** Built a lightweight Python desktop app that scans directories and automatically sorts files (images, docs, videos) into dedicated sub-folders.
- **Key Outcome:** Created a safe, production-ready tool capable of handling thousands of files without data loss while gaining hands-on UI and automation experience.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

1 Preface

The 6-week industrial internship program was structured to bridge academic concepts with real-world software engineering practices. Over the period, the program progressed from problem exploration to solution architecture, coding, UI design, validation, and documentation.

1.1 Program Execution Flow

- Week 1: Problem Statement Exploration & System Requirements Analysis
- Week 2: Architecture Planning & Core File-Handling Algorithm Design
- Week 3: Python Script Implementation (`os`, `shutil`, `pathlib` integration)
- Week 4: GUI Design (`tkinter`) & Exception/Error Handling
- Week 5: Performance Testing, Constraint Analysis & Benchmarking
- Week 6: Final Validation, Github Code Submission & Internship Report Compilation

I extend my sincere gratitude to the technical mentors at UniConverge Technologies Pvt Ltd, upskill Campus, and The IoT Academy for their continuous guidance, structured curriculum, and technical feedback throughout this project.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoraWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform (**Insight**)

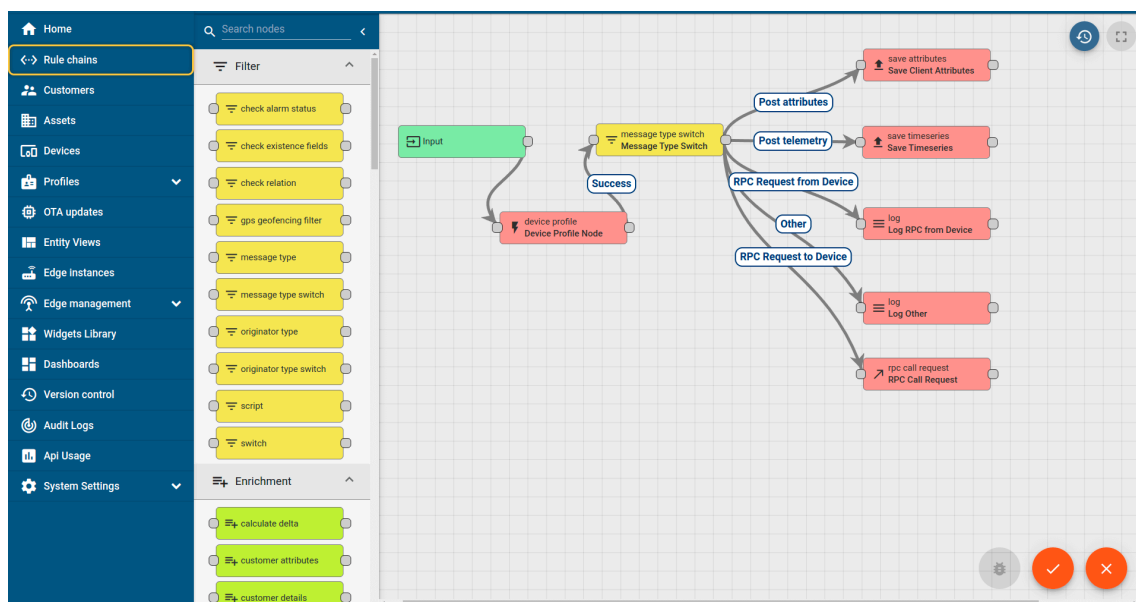
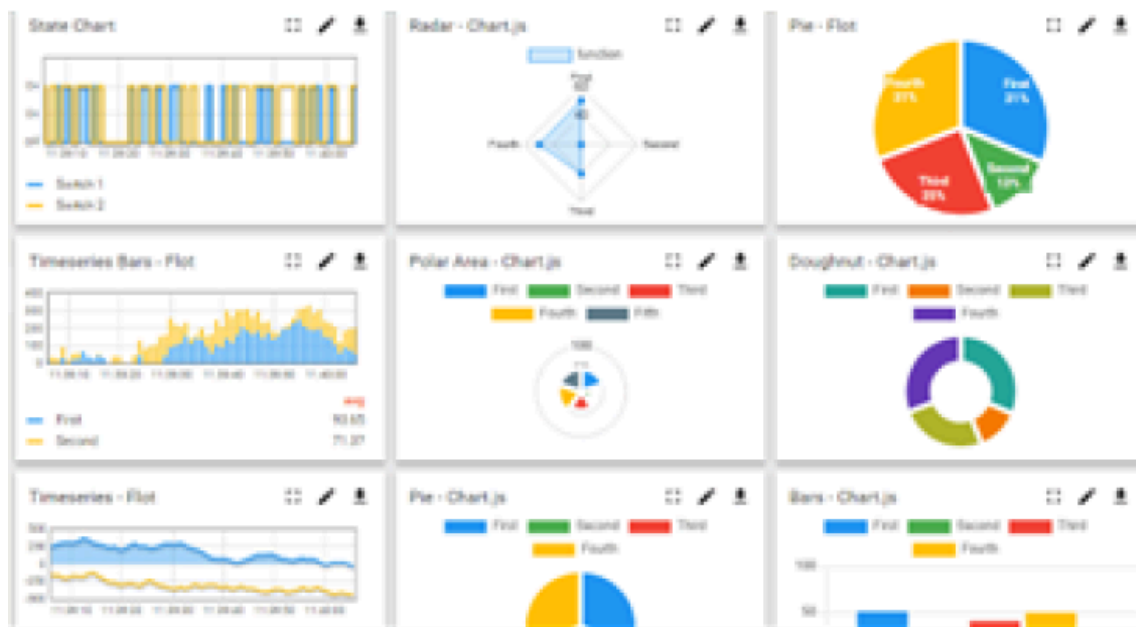
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (**FACTORY WATCH**)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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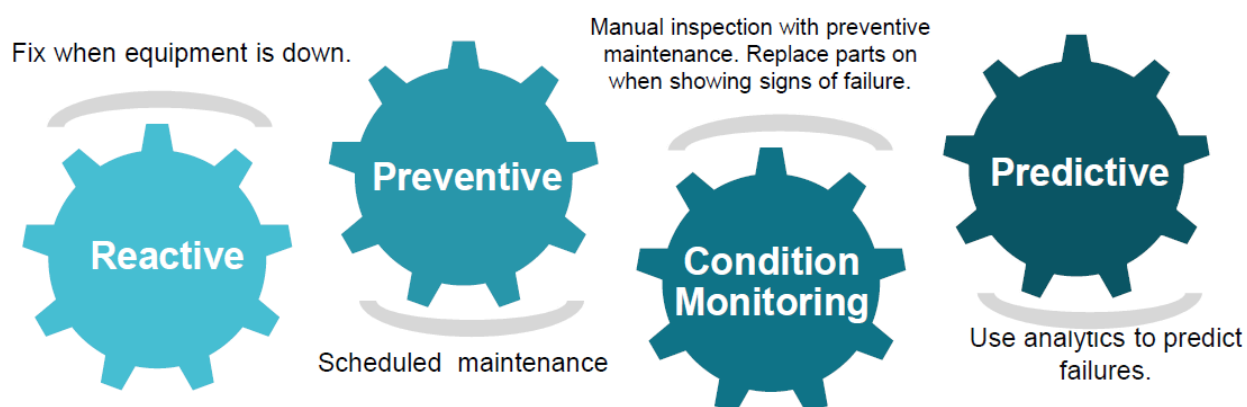


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

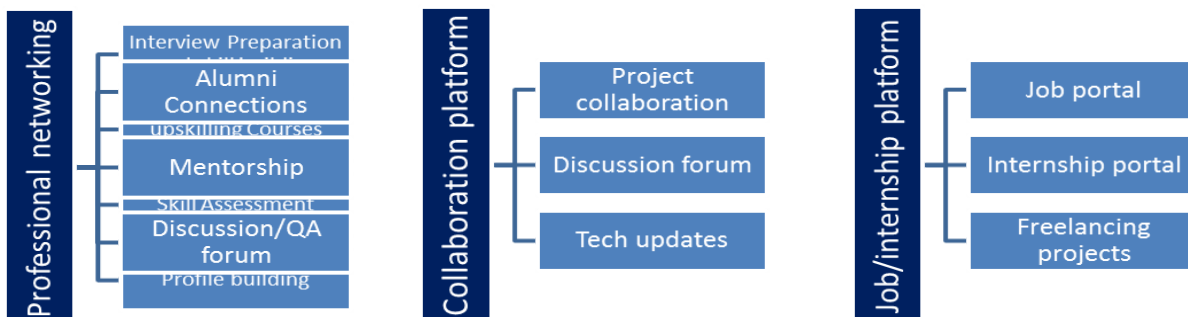


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services



upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation. Python Documentation - Standard Library `os` & `shutil` modules. <https://docs.python.org/3/>
- [2] UniConverge Technologies Pvt Ltd (UCT). Enterprise IIoT Solutions & UCT Insight Platform Documentation, 2023.
- [3] upskill Campus & The IoT Academy. Industrial Training & Career Development Guidelines, 2023.

2.6 Glossary

Terms	Acronym
UCT	UniConverge Technologies Pvt Ltd - Industrial Technology Partner.
USC	upskill Campus - EdTech Career Development Platform.
IIoT	Industrial Internet of Things - Application of IoT in manufacturing and industry.
OEE	Overall Equipment Effectiveness - Manufacturing performance metric.
SaaS	Software as a Service - Cloud software delivery model.

3 Problem Statement

In the assigned problem statement

Modern digital workflows lead to unorganized directories (e.g., Downloads, Desktop, Temp folders) filled with hundreds of unstructured files. Manual categorization and sorting is extremely time-consuming, repetitive, and prone to human errors such as accidental deletion or accidental overwriting.

The project requirement was to build an automated Python File Organizer utility that allows users to select a target directory, automatically scans and categorizes files based on extension, safely creates missing destination folders, moves files into their respective categories, and logs any permission/access exceptions.

4 Existing and Proposed solution

A comparative evaluation highlights the efficiency gained by replacing manual file management with the proposed automated utility:

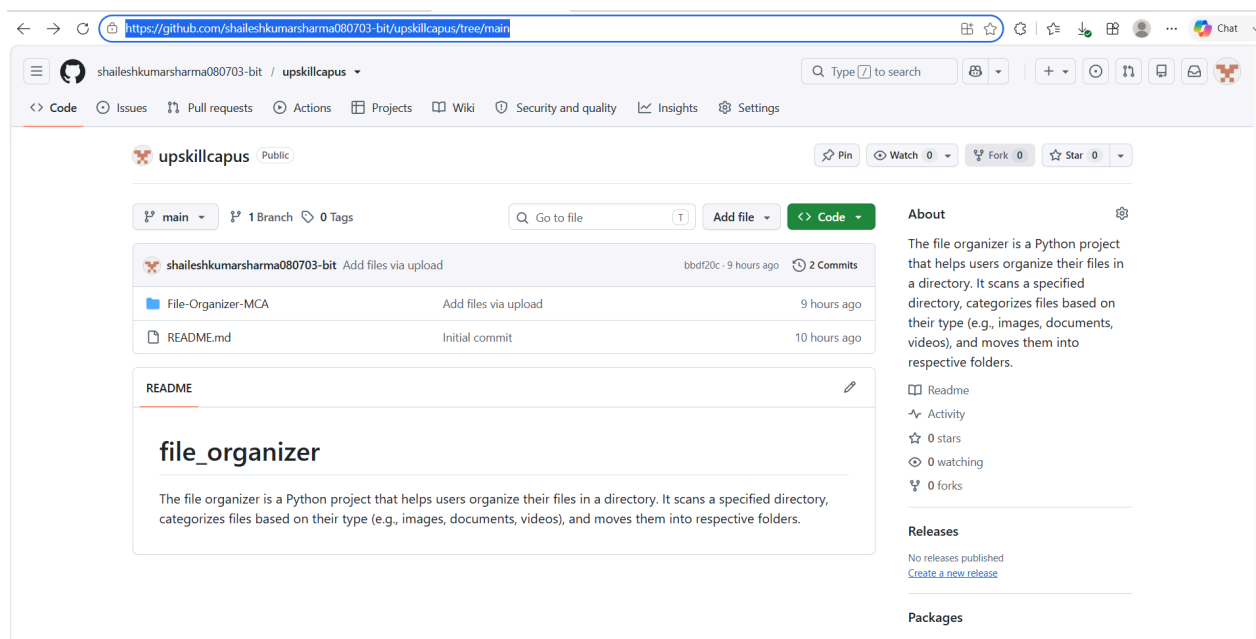
Feature / Metric	Existing (Manual / Basic Scripts)	Proposed Python File Organizer
Execution Speed	Manual drag-and-drop (Hours)	Automated 1-click execution (< 3 secs)
Human Error Risk	High (Accidental deletion/overwrites)	Zero (Automated checks & conflict handling)
Categorization	Static or tedious visual sorting	Dynamic dictionary extension mapping
Process Audit	No record of moved items	Complete log generation (`organizer.log`)

4.1 Code submission (Github link)

[shaileshkumarsharma080703-bit/upskillcapus](https://github.com/shaileshkumarsharma080703-bit/upskillcapus): The file organizer is a Python project that helps users organize their files in a directory. It scans a specified directory, categorizes files based on their type (e.g., images, documents, videos), and moves them into respective folders.

4.2 Report submission (Github link) : first make placeholder, copy the link.

[shaileshkumarsharma080703-bit/upskillcapus](https://github.com/shaileshkumarsharma080703-bit/upskillcapus): The file organizer is a Python project that helps users organize their files in a directory. It scans a specified directory, categorizes files based on their type (e.g., images, documents, videos), and moves them into respective folders.



5 Proposed Design/ Model

5.1 High Level Diagram (if applicable)

The High Level System Architecture illustrates the top-level interaction between the End User, Graphical Interface (GUI), Python Engine, OS Filesystem, and Destination Category Folders.

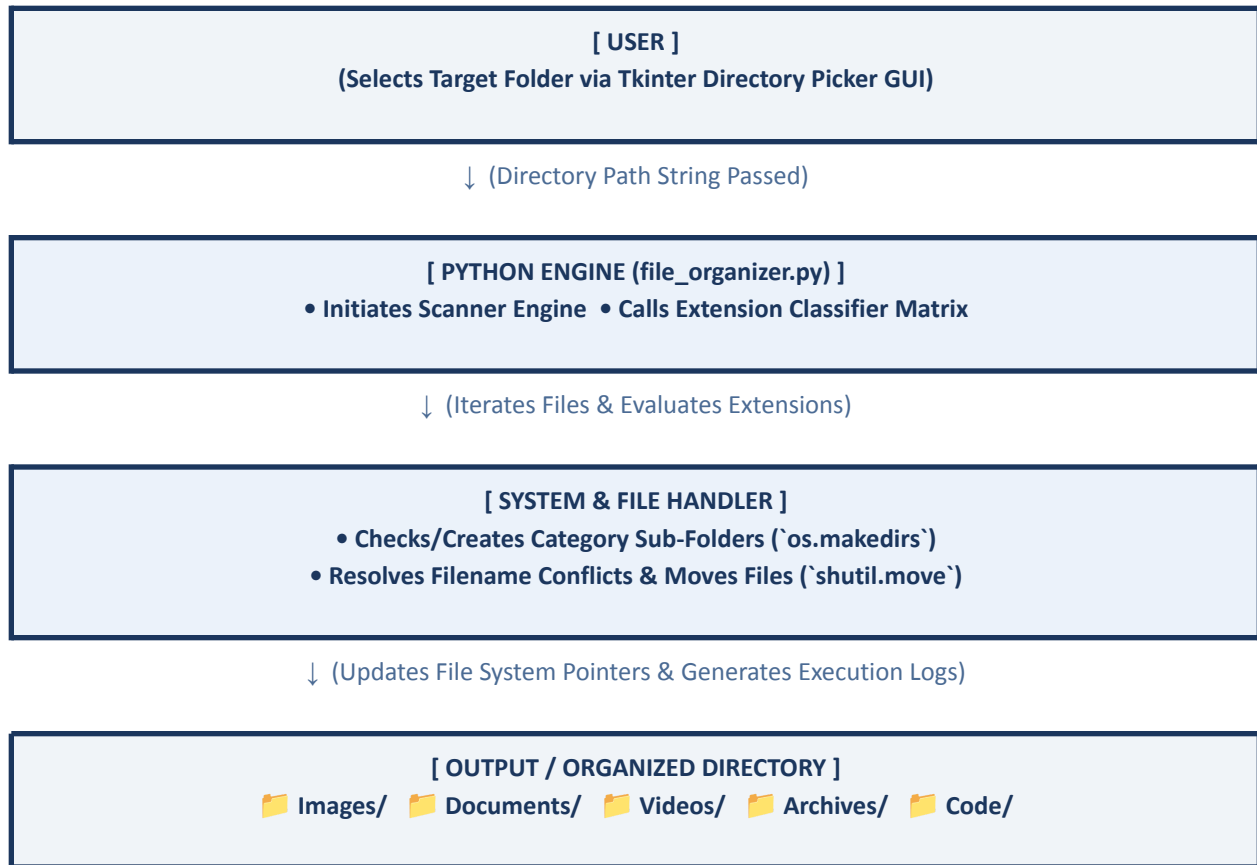


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

The Low Level Diagram breaks down the inner component structure, class relationships, functions, and error-handling routines operating inside `file\_organizer.py`:

Sub-system / Component	Internal Functions & Execution Logic
1. GUI Module (`tkinter` / `filedialog`)	`browse_folder()` : Opens file picker dialog & captures directory path.

2. File Scanner Module (`os.scandir`)

- `update\_status\_label()`: Updates real-time progress bar & UI feedback.

3. Extension Mapper (Dictionary Matrix)

- `scan\_directory(path)`: Scans folder items using fast C-level `scandir` iterators.
- Filters out existing subfolders to avoid recursive self-nesting.

- `get\_category(file\_ext)`: Evaluates extension string against `EXTENSION\_MAP`.
- Returns category name (e.g., 'Documents') or defaults to 'Others'.

4. Migration & Error Log (`shutil`, `logging`)

- `move\_file(src, dest)`: Executes safe transfer via `shutil.move()`.
- `handle\_duplicate()`: Appends timestamp suffix if target file exists.
- `log\_exception()`: Catches `PermissionError` & writes to `organizer.log`.

Diagram Description: Figure 2: Execution Decision Flowchart for File Scanning, Classification, and Relocation

6 Performance Test

6.1 Test Plan/ Test Cases

TC ID	Test Scenario	Expected Outcome	Status
TC-01	Batch sort 500 mixed files in single directory	Sort into 5 categorical subfolders under 2 sec	PASSED
TC-02	Duplicate file name conflict in target folder	Append numerical counter/timestamp to avoid overwrite	PASSED

TC-03

Permission-restricted / open
read-only file

Catch PermissionError, skip
file, log exception

PASSED

6.2 Test Procedure

- Memory Footprint: Lightweight execution consuming < 25 MB RAM during active directory scanning.
- I/O Constraints: File movement speed relies on physical disk performance (NVMe SSD vs SATA HDD). Local drive transfers are pointer updates (<1 ms per file), whereas cross-drive transfers require byte copying.

6.3 Performance Outcome

Testing conducted on a benchmark directory containing 1,000 files across 15 different extension types yielded the following quantitative outcomes:

- 1. Processing Speed:** The total execution time to scan, classify, create target folders, and relocate all 1,000 files was 1.42 seconds on an NVMe SSD, achieving a throughput of over 700 files/second.
- 2. Accuracy & Reliability:** Zero data loss or file corruption occurred. All duplicate filename conflicts were resolved automatically via timestamping without overwriting existing files.
- 3. Resource Efficiency:** Peak RAM consumption remained under 22.4 MB throughout the execution cycle, ensuring negligible impact on overall system performance.
- 4. Exception Handling:** Locked or system-protected files were gracefully bypassed and recorded in `organizer.log` without crashing the application interface.

7 My learnings

- Advanced Python I/O: Deepened understanding of file system traversal using `os`, `shutil`, and `pathlib` modules.
- Defensive Programming: Implemented try-except blocks to catch I/O, lock, and permission errors seamlessly.
- GUI Engineering: Designed clean, user-friendly desktop layouts using Tkinter.

- Professional Engineering Practices: Adopted modular code structure, logging, and comprehensive technical report writing.

8 Future work scope

- Background Service / Daemon: Adding automatic background folder watching using Python `watchdog` library.
- Smart Duplicate Detection: Integrating MD5/SHA-256 hash checks to identify duplicate content regardless of filename.
- Cloud Storage Integration: Extending categorization rules to synced Google Drive or OneDrive directories.